

Degly Sebastián Pava Pava

✉ deglypavapava@gmail.com 🌐 <https://deglypava.com> 🎓 IMLEX and Systems and Computer Engineer

From: August 2021
To: September 2023
(Currently in Japan)

Erasmus Mundus Japan – Master of Science in Imaging and Light in Extended Reality (IMLEX)

- Quality Board Member and Communications Representative
- Triple Degree Program (Msc. Optics, Image, Computer Vision, Msc. Engineering, Msc. Computer Science)
- Toyohashi University of Technology (Japan), KU Leuven (Belgium), University of Eastern Finland (Finland), University Jean Monnet Saint-Etienne (France)

Tech: Photonics | Realtime 3D Rendering | Computational Imaging | CUDA | OpenMP | Computer Graphics | Physical Optics | Machine Learning | ARKit | ARCore | VR/AR.

WORK EXPERIENCE

Company Owner

Omni Applications L.L.C. (Florida, U.S.A) – Lucid Dreams Ltd. (Bogotá, Colombia)

Tech: Project Management | ThreeJS | iOS | XCode | Unreal | SparkAR | 360 | LensStudio | Blender | Jira | Slack

From: January 2018
To: August 2021

Lead XR Developer | Hackspace for Innovation and Visualization in Education

Faculty of Medicine, The University of British Columbia, Vancouver, B.C. (Canada)

- Contributed to The HoloBrain and MR Manikin / OCS projects by engineering captivating 3D experiences integrated into a mixed reality environment using Unity 3D Engine. Presented these experiences at the *British Columbia Tech Summit* using a HoloLens v1 for an audience of over 200 participants and released them on Oculus Rift, HTC VIVE, and other Extended Reality devices.

Tech: Unity | Microsoft HoloLens 1&2 | OpenXR | Mixed Reality Toolkit (MRTK) | Vuforia | Blender | Neuroanatomy | Azure | Cloud Computing | Dependency Injection | Reactive Programming

From: February 2017
To: September 2019

Lead Developer and Engineer at Pontifical Xavierian University

Department of Industrial and Systems Engineering at Pontifical Xavierian University, Bogotá (Colombia)

- Created a 3D Material Requirements Planning (MRP) simulation system using NetLogo to optimize production planning, scheduling, and inventory control. Currently used in production classes to train students in MRP and scheduling concepts.

Tech: NetLogo | Anylogic | Linear Algebra | Microsoft Office Suite | Adobe Suite | Research Techniques

EDUCATION AND TRAINING

Certifications: VR / MR / XR Developer Meta Spark Creator AR Meta AR Developer AR & Video Streaming Services

Academic Recognitions:

- Bachelor of Systems and Computer Engineering**
 - Special Admission Program, best high school graduates of Colombia Scholarship at Universidad Nacional de Colombia
 - IronHacks – Purdue University Fellowship
 - Emerging Leaders of the Americas Program
- Master of Imaging and Light in Extended Realities**
 - IMLEX Full Tuition Award
 - ERASMUS Mundus Japan Scholarship
 - COLFUTURO PCB Scholarship
 - Manutech SLEIGHT Certificate Scholarship
 - BRMI Academic Mobility Award

Trained in multiple research groups:

Visual Perception and Cognition Laboratory
Hackspace for Innovation and Visualization in Education
Research Center for Open Digital Innovation
Grupo de investigación en procesos químicos y bioquímicos

Toyohashi University of Technology - Japan
The University of British Columbia - Canada
Purdue University – United States
Universidad Nacional de Colombia

From August 2021
To: September 2023

- Engineered a VR research experience, leveraging eye tracking technology and drawing from human psychology and HCI principles to analyze pupil responses in an immersive virtual environment.

From: January 2019
To: April 2021

- Designed and developed advanced neuroanatomy applications using MRI (Magnetic Resonance Imaging) and VR (Virtual Reality) technologies for medical research and education.

From: June 2018
To: December 2018

- Contributed to organizing the Gold IronHacks Purdue-UNAL Hackathon 2018. Also served as a judge, assessing participants' websites for the best implementation of Open Data based on specific attributes.

Digital skills	Extended Realities	Programming	Optics / Physics	Research and Tools
	Unity Blender ThreeJS	C# Python C++	Photonics Colorimetry	ImageJ3D Healthcare IT
	AWS Android UX design	JavaScript NetLogo Metal	Eye-Tracking Robotics	TensorFlow Photon Realtime
	Vuforia XR SDKs Arduino	Git Swift WebXR	Spectral Imaging	Mendeley Adobe Suite

PERSONAL SKILLS

Spanish Native proficiency

French Limited Working Proficiency

English Bilingual - IELTS (8.0) and iBT TOEFL (103 points)

Japanese Furigana / Basic Speaking and Writing